Python: exercises on iteration

This notebook was generated in student mode.

Exercise: Print Numbers from 1 to N

Exercise Description

Write a function that prints all numbers from 1 to N using a for loop and range.

Your solution

```
def print_numbers(n):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 1: Test with small number
print_numbers(5)
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Sum of First N Natural Numbers

Exercise Description

Write a function to calculate the sum of the first N natural numbers (1 + 2 + 3 + ... + N) using a while loop.

Your solution

```
def sum_natural_numbers(n):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 2: Test with another number
result = sum_natural_numbers(3)
```

```
expected = 1 + 2 + 3  # 6
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_natural_numbers(3) = {result}")
```

Test your solution

```
# Test case: Example from solution
result = sum_natural_numbers(5)
print(f"Sum of first 5 natural numbers: {result}")  # 15
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Print Even Numbers from 2 to 20

Exercise Description

Write a function that prints all even numbers between 2 and 20 using a for loop and range.

Your solution

```
def print_even_numbers():
    # Write your solution here
    pass
```

Test your solution

```
# Test case 2: Verify range and step
# Expected: prints 2, 4, 6, 8, 10, 12, 14, 16, 18, 20 (each on new line)
print_even_numbers()
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Count Down from N to 1 (If Even)

Exercise Description

Write a function to print numbers from N to 1 using a while loop, but only if N is even.

Your solution

```
def countdown_if_even(n):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 2: Test with odd number (should print nothing)
countdown_if_even(3)
```

Test your solution

```
# Test case 2 (assert)
countdown_if_even(6)  # Prints 6, 5, 4, 3, 2, 1
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Multiplication Table of 3

Exercise Description

Write a function that prints the multiplication table of 3 (from 3x1 to 3x10) using a for loop.

Your solution

```
def multiplication_table_3():
    # Write your solution here
    pass
```

Test your solution

```
# Test case 2: Verify format
# Expected: prints 3 x 1 = 3, 3 x 2 = 6, ..., 3 x 10 = 30
multiplication_table_3()
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Sum of First N Even Numbers

Exercise Description

Write a function to calculate the sum of the first N even numbers (2 + 4 + 6 + ...) using a while loop.

Your solution

```
def sum_first_n_even(n):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 2: Test with another number
result = sum_first_n_even(5)
expected = 2 + 4 + 6 + 8 + 10  # 30
assert result == expected, f"Expected {expected}, got {result}"
print(f"✓ sum_first_n_even(5) = {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Numbers Divisible by 5 (1 to 50)

Exercise Description

Write a function that prints all numbers divisible by 5 between 1 and 50 using a for loop and range.

Your solution

```
def print_divisible_by_5():
    # Write your solution here
    pass
```

Test your solution

```
# Test case 2: Verify range
# Expected: prints 5, 10, 15, 20, 25, 30, 35, 40, 45, 50
print_divisible_by_5()
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Factorial of 5

Exercise Description

Write a function to find the factorial of 5 using a while loop.

Your solution

```
def factorial_5():
    # Write your solution here
    pass
```

Test your solution

```
# Test case 2: Verify result
result = factorial_5()
```

```
assert result == 120, f"Expected 120, got {result}"
print(f"✓ Factorial of 5 is {result}")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Sum of Odd Numbers (1 to 20)

Exercise Description

Write a function to calculate the sum of all odd numbers between 1 and 20 using a for loop.

Your solution

```
def sum_odd_numbers():
    # Write your solution here
    pass
```

Test your solution

```
# Test case 2: Verify result
result = sum_odd_numbers()
assert result == 100, f"Expected 100, got {result}"
print(f"✓ Sum of odd numbers 1-20 is {result}")
```

Test your solution

```
# Test case: Example from solution
result = sum_odd_numbers()
print(f"Sum of odd numbers 1-20: {result}")  # 100
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Numbers Divisible by K (1 to N)

Exercise Description

Write a function to print numbers between 1 and N that are divisible by K using a for loop and range.

Your solution

```
def print_divisible_by_k(n, k):
    # Write your solution here
    pass
```

Test your solution

```
# Test case 2: Test with n=20, k=4
# Expected: prints 4, 8, 12, 16, 20
print_divisible_by_k(20, 4)
```

Test your solution

```
# Test case: Example from solution
print_divisible_by_k(20, 3)  # Prints 3, 6, 9, 12, 15, 18
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Count Digits in a Number

Exercise Description

Write a function that counts how many digits are in a positive integer.

Your solution

```
def count_digits(num):
    """Return the number of digits in positive integer num."""
    # Write your solution here
    pass
```

Test your solution

```
assert count_digits(123456) == 6
assert count_digits(7) == 1
assert count_digits(10) == 2
print("✓ count_digits tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the

Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Cubes from 1 to 5

Exercise Description

Write a function that returns a list with the cubes of numbers from 1 to 5 using a for loop.

Your solution

```
def cubes_1_to_5():
    # Write your solution here
    pass
```

Test your solution

```
assert cubes_1_to_5() == [1, 8, 27, 64, 125]
print("✓ cubes_1_to_5 tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Product of Numbers from 1 to N

Exercise Description

Write a function that returns the product of numbers from 1 to N using a for loop.

Your solution

```
def product_1_to_n(n):
    # Write your solution here
    pass
```

Test your solution

```
assert product_1_to_n(1) == 1
assert product_1_to_n(3) == 6
assert product_1_to_n(5) == 120
print("✓ product_1_to_n tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Odd Numbers from 1 to 15

Exercise Description

Write a function that returns all odd numbers from 1 to 15 using a while loop.

Your solution

```
def odd_1_to_15():
    # Write your solution here
    pass
```

Test your solution

```
assert odd_1_to_15() == [1, 3, 5, 7, 9, 11, 13, 15]
print("✓ odd_1_to_15 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Print Numbers from 5 to 1

Exercise Description

Write a function that returns numbers from 5 to 1 using a for loop with range.

Your solution

```
def countdown_5_to_1():
    # Write your solution here
    pass
```

Test your solution

```
assert countdown_5_to_1() == [5, 4, 3, 2, 1]
print("✓ countdown_5_to_1 tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Sum of the First 7 Odd Numbers

Exercise Description

Write a function to find the sum of the first 7 odd numbers using a while loop.

Your solution

```
def sum_first_7_odds():
    # Write your solution here
    pass
```

Test your solution

```
assert sum_first_7_odds() == 49
print("✓ sum_first_7_odds tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Repeat Until "stop"

Exercise Description

Write a function that, given a sequence of strings, counts how many entries occur until the string "stop" appears.

Your solution

```
def until_stop_count(inputs):
    """Return the number of entries until "stop" appears (case-sensitive)."""
    # Write your solution here
    pass
```

Test your solution

```
assert until_stop_count(["a", "b", "stop", "x"]) == 2
assert until_stop_count(["stop"]) == 0
assert until_stop_count(["hello", "world"]) == 2
print("✓ until_stop_count tests passed")
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Repeat String as Many Times as Its Length

Exercise Description

Write a function that takes a string and returns a list with the string repeated as many times as its length.

Your solution

```
def repeat_string_len_times(s):
    # Write your solution here
    pass
```

Test your solution

```
assert repeat_string_len_times("abc") == ["abc", "abc", "abc"]
assert repeat_string_len_times("") == []
print("✓ repeat_string_len_times tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.
```

Exercise: Ask for a Password

Exercise Description

Write a function that, given a sequence of password attempts, returns the number of attempts needed to enter the correct password ("password123"). Return -1 if never entered.

Your solution

```
def password_attempts(attempts, correct="password123"):
    # Write your solution here
    pass
```

Test your solution

```
assert password_attempts(["x", "y", "password123"]) == 3
assert password_attempts(["password123"]) == 1
assert password_attempts(["a", "b"]) == -1
print("✓ password_attempts tests passed")
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.